

AMAL JYOTHI
COLLEGE OF ENGINEERING
AUTONOMOUS
KANJIRAPPALLY

PORTIDE : MARITIME PORT OPERATIONS INTELLIGENCE PLATFORM

24MCAR295 - Research Project/Software

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ABSTRACT

Maritime ports play a vital role in global trade by facilitating the movement of vessels, cargo, containers, and logistics resources across international supply chains. As global trade continues to increase, ports face significant challenges such as vessel congestion, inefficient berth utilization, delays in cargo handling, poor resource allocation, fragmented operational management, and limited coordination among multiple stakeholders. Existing port management systems primarily focus on operational record management and monitoring while providing limited support for intelligent planning, predictive analytics, and integrated decision-making. The absence of a centralized platform often leads to inefficient resource utilization, communication gaps between stakeholders, operational delays, and difficulty in managing complex port activities efficiently.

The proposed Portide: Maritime Port Operations Intelligence Platform is an enterprise-grade web application designed to centralize and streamline maritime port operations within a unified digital platform. The system enables Port Administrators, Shipping Companies, Port Operations Officers, Customs Officers, Terminal Operators, Logistics Operators, and Gate Operations Officers to efficiently manage vessel scheduling, berth allocation, cargo and container operations, customs coordination, resource allocation, gate operations, workflow management, operational notifications, reporting, and intelligent decision support through a single integrated system.

FUNCTIONAL MODULES

PHASE 1

Complete Enterprise-grade Maritime Port Operations Intelligence Platform

Module 1 – User Authentication & Role Management

- Port Administrator
- Shipping Company / Shipping Agent
- Port Operations Officer
- Customs Officer
- Terminal & Logistics Operator
- Gate & Truck Operations Officer
- Secure authentication using JWT
- Role-Based Access Control

Module 2 – Vessel Scheduling & Arrival Management

- Vessel Registration
- Arrival Scheduling
- ETA Management
- Departure Scheduling
- Vessel Status Tracking

Module 3 – Smart Berth & Dock Management

- Berth Availability Monitoring
- Intelligent Berth Allocation
- Dock Occupancy Tracking
- Waiting Vessel Queue
- Berth Utilization Dashboard
- Vessel Berth Shifting Management

Module 4 – Cargo & Container Management

- Cargo Registration
- Container Tracking
- Loading & Unloading Management
- Yard Allocation
- Container Status Monitoring

Module 5 – Customs & Documentation Management

- Document Upload
- Cargo Verification
- Customs Inspection Workflow
- Clearance Approval
- Compliance Monitoring

Module 6 – Resource Management

- Crane Allocation
- Workforce Scheduling
- Equipment Tracking
- Resource Utilization Monitoring
- Maintenance Scheduling

Module 7 – Gate & Truck Operations Management

- Truck Registration
- Appointment Scheduling

- Digital Gate Pass Generation
- Gate Entry & Exit Monitoring
- Container Pickup Tracking

Module 8 – Workflow Automation & Notification Hub

- Vessel Arrival Alerts
- Berth Assignment Notifications
- Customs Clearance Alerts
- Operational Delay Notifications
- Truck Appointment Notifications
- Resource Availability Alerts

Module 9 – Analytics & Reporting

- Operational Dashboard
- Vessel Statistics
- Cargo Statistics
- Resource Utilization Reports
- Container Flow Reports
- KPI Dashboard

PHASE 2

AI-powered Maritime Port Operations Intelligence Platform

Module 1 – Port Congestion Prediction

Purpose

Predict future congestion levels using historical vessel movement, berth occupancy, weather conditions, cargo volume, and operational data to support proactive planning.

Features

- Port congestion prediction
- Peak traffic detection
- Vessel waiting time estimation
- Congestion risk alerts
- Operational trend analysis

Module 2 – Vessel Turnaround Time Prediction

Purpose

Estimate vessel turnaround time by analyzing cargo operations, berth allocation, customs processing, workforce, and equipment availability.

Features

- Turnaround time prediction
- Delay prediction
- ETA/ETD estimation
- Operational efficiency analysis
- Performance insights

Module 3 – Intelligent Berth Recommendation

Purpose

Recommend the most suitable berth based on vessel characteristics, berth availability, cargo type, operational priorities, and equipment availability.
Features

- Intelligent berth recommendation
- Dock assignment optimization
- Berth utilization analysis
- Conflict detection
- Explainable berth allocation

USER MODULES

1. Port Administrator

Functionalities

- Manage users and roles
- Configure port operations
- Monitor dashboards
- View reports
- Manage the overall platform

2. Shipping Company / Shipping Agent

Functionalities

- Register vessels
- Submit arrival requests
- Upload cargo manifests
- Track vessel schedules
- Receive operational notifications

3. Port Operations Officer

Functionalities

- Schedule vessel arrivals
- Allocate berths
- Assign cranes and equipment
- Monitor vessel turnaround
- Track operational delays

4. Customs Officer

Functionalities

- Verify cargo documents
- Conduct inspections
- Approve customs clearance
- Monitor compliance
- Generate customs reports

5. Terminal & Logistics Operator

Functionalities

- Manage container yard operations
- Allocate storage locations
- Monitor cargo movement
- Coordinate loading and unloading
- Track container delivery

6. Gate & Truck Operations Officer

Functionalities

- Register trucks
- Schedule appointments
- Generate digital gate passes
- Monitor gate operations
- Track container pickup and delivery